Consumption-Savings with Markov Income Risk

Group 2

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Part I: Two-Period Model

An agent begins period 1 with cash-on-hand x_1 and current income state y_1. There are two periods, $t = 1, 2$. The gross interest rate is $R > 0$. Income next period, y_2, is random and depends on current income through a discrete-state Markov process.

Let the income state space be

$$\mathcal{Y} = \{y_1, ..., y_N\},$$

with transition probabilities

$$\Pi(y, y') = \Pr(y_{t+1} = y' | y_t = y).$$

At date 1 the current state y_1 is known, but y_2 is uncertain. Preferences are

$$u(c_1) + \beta \mathbb{E}_1[u(c_2)], \quad u(c) = \begin{cases} \frac{c^{1-\gamma}}{1-\gamma}, & \gamma \neq 1, \\ \log c, & \gamma = 1. \end{cases}$$

The period-1 budget constraint is

$$c_1 + a_2 = x_1,$$

and period-2 consumption is

$$c_2 = y_2 + Ra_2.$$

The borrowing constraint is

$$a_2 \geq \underline{a}.$$

Questions

1.1. Write the agent's period-1 problem using a_2 as the choice variable.

1.2. Derive the Euler equation for an interior solution.

1.3. Explain in words how the Markov structure differs from the iid case studied in class.

1.4. Derive the threshold level of cash-on-hand, call it $x_1^*(y_1)$, at which the agent is just willing to choose the borrowing-constraint level $a_2 = \underline{a}$.

1.5. Explain how $x_1^*(y_1)$ depends on the current income state through the transition probabilities.

Part II: Infinite-Horizon Model

Now consider the infinite-horizon version. Time is discrete, $t = 0, 1, 2, \ldots$. Each period the agent begins with assets a and current income state $y \in \mathcal{Y}$. Income follows the Markov process with transition matrix Π.

The gross interest rate is R, the discount factor is β, and the borrowing constraint is $a' \geq \underline{a}$. The period budget constraint is $c + a' = y + Ra$.

The Bellman equation is

$$V(a, y) = \max_{a' \geq \underline{a}} \{u(c) + \beta \sum_{y' \in \mathcal{Y}} \Pi(y, y') V(a', y')\}$$

subject to

$$c = y + Ra - a'.$$

For an interior solution, the Euler equation is

$$u'(c) = \beta R \sum_{y' \in \mathcal{Y}} \Pi(y, y') u'(c'(a', y')).$$

Use the file `IntroEndogenousGrid_Infinite_Uncertainty.m` as your starting point. Your task is to modify it so that uncertainty comes from a discrete-state Markov income process rather than from an iid shock.

Required Tasks

2.1. Modify the MATLAB code so that it solves the infinite-horizon model with Markov income risk using EGM.

2.2. Plot the consumption policy $c(a, y)$ for at least two income states.

2.3. Plot the savings policy $a'(a, y)$ for at least two income states.

2.4. Compute the stationary distribution induced by the converged policy rule.

2.5. Report a small set of stationary moments, such as average assets, average consumption, and the share of households at the borrowing constraint.

2.6. Study a simple numerical experiment that changes the persistence of the income process while keeping average income fixed. Explain how the policy functions and stationary moments change.

Optional Extension

If you would like to go a bit further, you may compare your EGM solution to a coarser linear-spline value function iteration solution and comment briefly on the differences in speed, smoothness, and implementation.

Answer:

1 Part I: Two-Period Model

1.1 Consumption-Savings with Markov Income Risk

We have $c_1 + a_2 = x_1 \quad \implies c_1 = x_1 - a_2$ and $c_2 = y_2 + Ra_2$ from question.

We can now substitute these values into our preferences:

$$u(c_1) + \beta E_1[u(c_2)] = u(x_1 - a_2) + \beta E_1[u(y_2 + Ra_2)]$$

And so the problem is:

$$\max_{a_2 \geq \underline{a}} \{u(x_1 - a_2) + \beta E_1[u(y_2 + Ra_2)]\}$$

The key difference in this problem is that next-period income is Markov, the expectation can be written as a sum over future income states, as given in question:

$$E_1[u(y_2 + Ra_2)] = \sum_{y' \in Y} \Pi(y_1, y')u(y' + Ra_2)$$

Therefore an equivalent expression is the following:

$$\max_{a_2 \geq \underline{a}} \{u(x_1 - a_2) + \beta \sum_{y' \in Y} \Pi(y_1, y')u(y' + Ra_2)\}$$

1.2 Derive the Euler equation for an interior solution.

Starting from the objective in 1.1,

$$\max_{a_2 \geq \underline{a}} \{u(x_1 - a_2) + \beta \mathbb{E}_1[u(y_2 + Ra_2)]\}.$$

If the optimum is interior, so that $a_2 > \underline{a}$, the first-order condition is

$$-u'(x_1 - a_2) + \beta R \mathbb{E}_1[u'(y_2 + Ra_2)] = 0.$$

Giving us the desired Euler equation:

$$u'(x_1 - a_2) = \beta R \mathbb{E}_1[u'(y_2 + Ra_2)].$$

1.3 Explain in words how the Markov structure differs from the iid case studied in class.

By 1.2, we can see the difference mathematically by expanding the expectation. In the iid case, the law of future income does not depend on the current state y_1, so

$$E_1[u'(c_2)] = \sum_{y' \in Y} \pi(y')u'(y' + Ra_2),$$

where π is the common distribution of next-period income. Thus the expectation is the same regardless of the current income state.

In the Markov case,

$$E_1[u'(c_2)] = \sum_{y' \in Y} \Pi(y_1, y')u'(y' + Ra_2),$$

so the expectation depends on y_1 through the transition probabilities. In general probability theory, independence is the stronger restriction, since the Markov property still allows dependence across time. But in this model the Markov case gives the consumer more information at date 1, because the current state y_1 helps determine the distribution of tomorrow's income. Thus, unlike in the iid case, the current income state has predictive content for future income.

1.4 Derive the threshold level of cash-on-hand, call it $x_1^*(y_1)$, at which the agent is just willing to choose the borrowing-constraint level $a2 = a$.

Define the agent's objective as

$$W(a_2; x_1, y_1) = u(x_1 - a_2) + \beta \sum_{y' \in Y} \Pi(y_1, y')\, u(y' + Ra_2).$$

Because utility is concave, the objective is concave in a_2, so the optimum is either:

- at the borrowing limit $a_2 = \underline{a}$, or
- at an interior point satisfying the first-order condition.

Differentiating with respect to a_2 gives

$$\frac{\partial W}{\partial a_2} = -u'(x_1 - a_2) + \beta R \sum_{y' \in Y} \Pi(y_1, y')\, u'(y' + Ra_2).$$

Hence, for an interior solution, the Euler equation is

$$u'(x_1 - a_2) = \beta R \sum_{y' \in Y} \Pi(y_1, y')\, u'(y' + Ra_2).$$

The threshold $x_1^*(y_1)$ is defined as the level of cash-on-hand at which the agent is just willing to choose the borrowing limit $a_2 = \underline{a}$. At this cutoff, the derivative of the objective at the borrowing limit is zero:

$$\left.\frac{\partial W}{\partial a_2}\right|_{a_2 = \underline{a},\ x_1 = x_1^*(y_1)} = 0.$$

Substituting into the derivative gives

$$-u'(x_1^*(y_1) - \underline{a}) + \beta R \sum_{y' \in Y} \Pi(y_1, y')\, u'(y' + R\underline{a}) = 0.$$

Rearranging, the threshold condition is

$$u'(x_1^*(y_1) - \underline{a}) = \beta R \sum_{y' \in Y} \Pi(y_1, y')\, u'(y' + R\underline{a}).$$

This equation implicitly defines $x_1^*(y_1)$. It is the cutoff level of cash-on-hand such that:

$$x_1 < x_1^*(y_1) \quad \Rightarrow \quad a_2 = \underline{a},$$

while

$$x_1 > x_1^*(y_1) \quad \Rightarrow \quad a_2 > \underline{a}.$$

If utility is CRRA with $\gamma \neq 1$, so that $u'(c) = c^{-\gamma}$, then the threshold becomes

$$(x_1^*(y_1) - \underline{a})^{-\gamma} = \beta R \sum_{y' \in Y} \Pi(y_1, y')\, (y' + R\underline{a})^{-\gamma},$$

and therefore

$$x_1^*(y_1) = \underline{a} + \left[\beta R \sum_{y' \in Y} \Pi(y_1, y')\, (y' + R\underline{a})^{-\gamma}\right]^{-1/\gamma}.$$

If $\gamma = 1$, so utility is logarithmic and $u'(c) = 1/c$, then

$$\frac{1}{x_1^*(y_1) - \underline{a}} = \beta R \sum_{y' \in Y} \Pi(y_1, y')\, \frac{1}{y' + R\underline{a}},$$

which implies

$$x_1^*(y_1) = \underline{a} + \left[\beta R \sum_{y' \in Y} \Pi(y_1, y')\, \frac{1}{y' + R\underline{a}}\right]^{-1}.$$

1.5 Explain how $x_1^*(y_1)$ depends on the current income state through the transition probabilities.

The threshold $x_1^*(y_1)$ depends on the current income state y_1 through the transition probabilities $\Pi(y_1, y')$. This can be seen directly from the threshold condition:

$$u'(x_1^*(y_1) - \underline{a}) = \beta R \sum_{y' \in Y} \Pi(y_1, y')\, u'(y' + R\underline{a}).$$

The right-hand side is the discounted expected marginal utility of future consumption when the agent chooses the borrowing-limit asset position $a_2 = \underline{a}$. Since the probabilities $\Pi(y_1, y')$ depend on the current state y_1, the expected marginal utility term also depends on y_1.

The economic intuition is as follows. Suppose income is positively persistent.

- If current income y_1 is low, then low future income states are more likely. At the borrowing limit, this implies lower expected future consumption, so expected future marginal utility is higher. Saving therefore becomes more valuable today. As a result, the household is willing to move away from the borrowing limit at a lower level of current cash-on-hand, so $x_1^*(y_1)$ is lower.
- If current income y_1 is high, then high future income states are more likely. At the borrowing limit, this implies higher expected future consumption, so expected future marginal utility is lower. The precautionary saving motive is therefore weaker. In this case, the household is willing to remain at the borrowing limit for a wider range of cash-on-hand levels, so $x_1^*(y_1)$ is higher.

Thus, with positive persistence, the threshold is typically lower in low-income states and higher in high-income states.

This highlights the difference between the Markov case and the iid case. Under iid income risk, the distribution of y_2 does not depend on y_1, so the expectation

$$\sum_{y' \in Y} \Pi(y_1, y')\, u'(y' + R\underline{a})$$

would be the same for all current states. In that case, the threshold $x_1^*(y_1)$ would not vary with y_1. Under Markov income risk, however, the threshold becomes state-dependent because the current income state changes the conditional distribution of future income.

Therefore, persistence affects saving behaviour by shifting the borrowing-threshold cutoff across current income states. Two households with the same current cash-on-hand may make different saving choices if they face different current income states, because those states imply different expectations about future income.

2 Part II: Infinite-Horizon Model

2.1

Adaption of the EGM to incorporate Markov income risk required modification of the Euler equation; specifically, how expectations are computed in the equation. In the original format, expected future marginal utility is determined by independent income draws. Whereas in the adapted version with Markov income, future income is dependent on current state via the transition matrix.

Incorporating this element, the Euler equation becomes

$$u'(c_t) = \beta R \sum_{y'} \Pi(y, y') u'(c_t + 1(a', y'))$$

Within the code, the critical modification is the computation of expected marginal utility. For each current income state y, expected marginal utility is calculated as a probability-weighted sum across all possible future income states y'. These weights come directly from the transition matrix, incorporating income persistence in the model.

The remaining components of the EGM procedure remain the same as the original version, in that; a grid is fixed for next-period assets a'; the Euler equation is used to compute current consumption; an endogenous grid is constructed for cash-on-hand; and policy functions are interpolated back onto the exogenous asset grid. The combinations of these elements, along with the Markov adaptions allow the model to capture persistent income risk, which in turn allows for realistic precautionary savings behavior.

2.2

In the graphical results for the Consumption Policy Function, c(a,y), we see the impacts of income & asset levels, wealth effect, and borrowing constraint. The distinction between high- and low-income groups is evident across the graph, as at any asset level consumption is higher for the high-income group. This is a consequence of the wealth effect; where higher assets result in a larger amount of available resources and thus higher consumption. For the high-income group, cash-on-hand is larger, as are the expectations of future income; a result of the persistence introduced by the Markov process.

While there is a clear distinction between income groups, both groups' consumption levels vary dramatically depending on asset level. When assets are near zero, or negative, actions are closely tied to the borrowing constraint. For the low-income group, this means very little consumption, as both income and cash-on-hand are low and the agent can borrow no further. However, as assets rise, consumption increases drastically, as the burden of debt repayment minimizes. This is contrasted by the high-income group, where even at negative assets, consumption can remain high due to high income levels. Consumption for this group then declines as assets approach zero as agents begin to save more in response to future uncertainty; a consequence of the transition from a constraint-dominated region to an optimization-driven one. For both high- and low-income groups, as assets increase beyond zero, their respective consumption functions increase and become smoother, reflecting standard consumption smoothing behavior, as expected by the Euler equation.

The graphical results from a'(a,y), the Saving Policy Function, serve to reinforce the conclusions drawn from the Consumption Policy; namely the importance of consumption smoothing, borrowing constraints, and precautionary actions under persistent uncertainty. Generally, the results demonstrate how agents allocate resources between their current consumption and future assets. This means that for both income groups, savings increase as assets do, as wealthier agents have greater capacity to save. Likewise, as assets are low or negative, agents prioritize consumption, resulting in low savings.

As with the Consumption Policy Function, the high-income group saves more across asset levels. This can be attributed to their greater current resource amount, as well as income persistence which encourages expectations of higher future income. Conversely, the low-income group saves less as they experience tighter resource constraints.

The incorporation of Markov income risk in the model also plays a critical role in the results as it encourages precautionary saving motives. For the low-income group, this translates to an expectation of future low income, meaning a greater importance of saving to sustain themselves. Likewise, the high-income group are also encouraged to save to smooth consumption overtime.

2.3 Introduction

We follow the assignment details and define the bellman model in the following manner.

Let $V(a, y)$ denote the value of starting a period with assets a and current income state y. Then the infinite-horizon problem can be written recursively as

$$V(a, y) = \max_{a' \geq \underline{a}} \left\{ u(y + Ra - a') + \beta \sum_{y' \in Y} \Pi(y, y') V(a', y') \right\}.$$

Here $u(y + Ra - a')$ is current utility, since current consumption is given by

$$c = y + Ra - a',$$

and the second term is the discounted expected continuation value. The Markov structure enters through the transition probabilities $\Pi(y, y')$, so the continuation value depends on the current income state y.

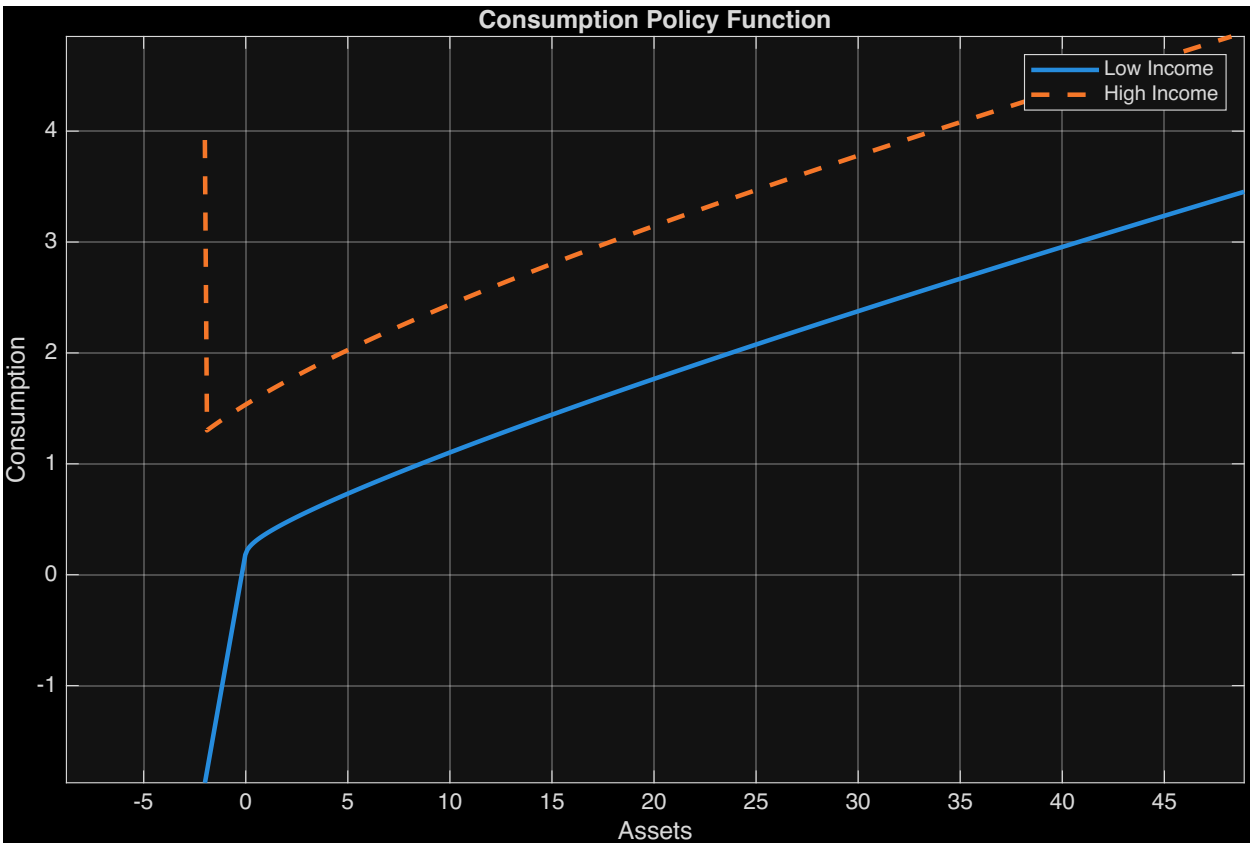

Figure 1: Consumption Policy Function

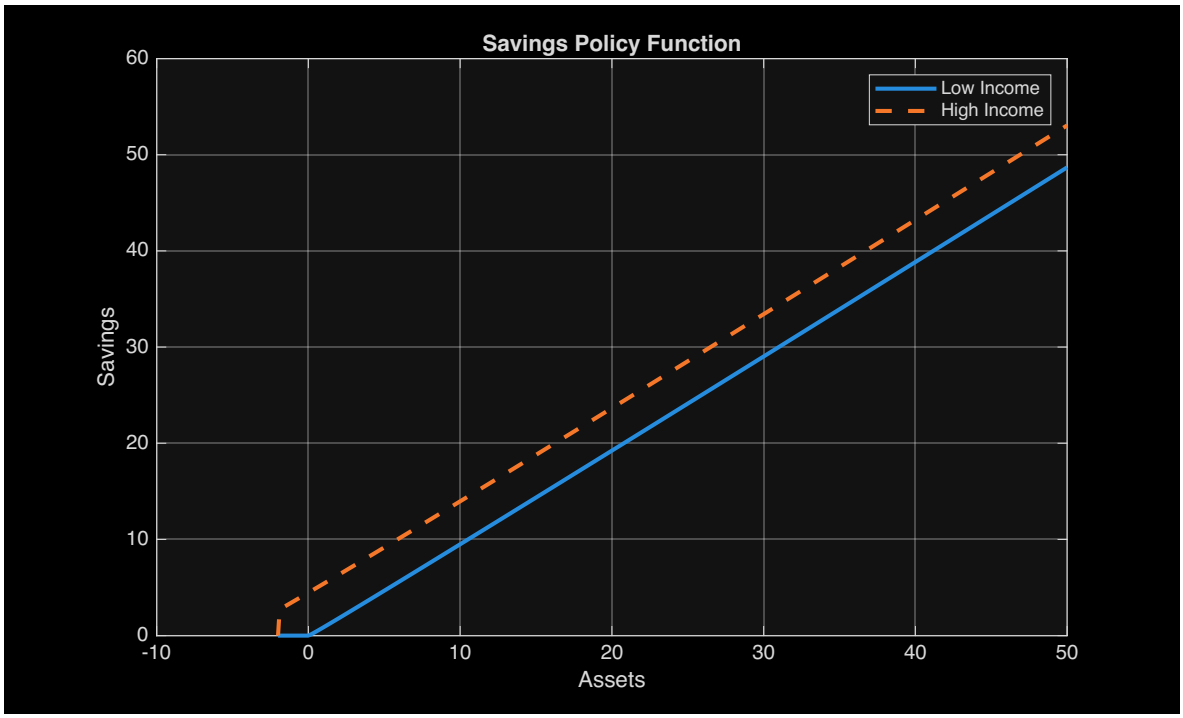

Figure 2: Saving Policy Function

The Markov structure carries directly into the infinite-horizon model through the continuation value. In each period, the current income state y determines the distribution of next period's income state y' through the transition matrix $\Pi(y, y')$. As a result, the expected continuation value is

$$\sum_{y' \in Y} \Pi(y, y') V(a', y'),$$

so it depends on the current state y. Thus, unlike in the iid case, the current income state is part of the state variable, and both the value function and the optimal policy rules are state-dependent.

As discussed above, we now have additional information in the Markov setting. As a result, EGM must be implemented state by state. For each income state y, one fixes a grid for next-period assets a', computes expected future marginal utility using the corresponding row of Π, inverts the Euler equation to recover current consumption, and then uses the budget constraint to obtain the implied current asset level.

2.4 The Stationary Distribution

The stationary distribution represents the long-run steady state of the economy, where the fraction of households at each asset level remains constant over time despite individual movements across income states[cite: 49, 68]. To compute this distribution, we utilized a Monte Carlo simulation involving 10,000 agents over a horizon of 500 periods.

The resulting distribution demonstrates a unimodal "hump" shape, centered significantly to the right of the borrowing constraint. This shift reflects the high degree of self-insurance achievable in this environment. The lack of a mass at the borrowing limit ($\underline{a} = -2$) suggests that agents have accumulated sufficient wealth buffers to avoid hitting the constraint during typical low-income spells.

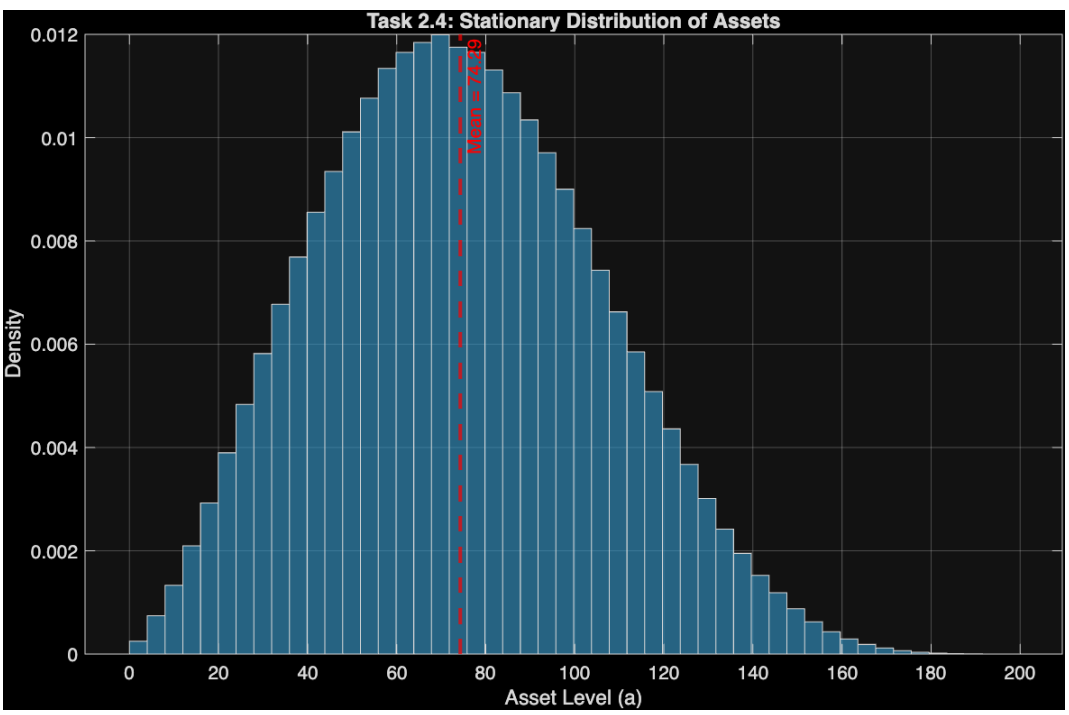

Figure 3: Task 2.4: Stationary Distribution of Assets ($\bar{a} = 74.48$)

2.5 Stationary Moments

Following the simulation of the converged policy rules, we extracted key stationary moments to summarize the long-run aggregate state of the economy[cite: 50, 51]. The results are reported in Table 1.

Table 1: Stationary Moments Summary

Moment	Value
Average Assets	74.4824
Average Consumption	5.5149
Share at Borrowing Constraint	0.00%

The average asset level of 74.48 is high relative to the income states, signaling a robust precautionary saving motive. Furthermore, the average consumption of 5.51 indicates that agents are successfully smoothing consumption by utilizing both labor income and the 4% interest return on their substantial asset holdings.

2.6 Numerical Experiment on Persistence

We conducted a numerical experiment to study how the persistence of the Markov income process affects saving behavior while keeping the average income fixed[cite: 52, 53, 69].

Changes in Policy Functions

As income persistence increases (higher diagonal elements in Π), the risk of a long-duration "low income" spell grows. This strengthens the precautionary motive, causing the savings policy $a'(a, y)$ to shift upward and the consumption policy $c(a, y)$ to shift downward. Agents reduce current consumption to build larger buffers against persistent negative shocks.

Changes in Stationary Moments

In the high-persistence baseline (0.85), average assets reached 74.48. In contrast, reducing persistence (making income shocks more transitory) leads to a dramatic reduction in average assets. When shocks are less persistent, the need for self-insurance decreases, allowing agents to maintain lower asset levels without significantly increasing the frequency with which the borrowing constraint binds.

3 Optional Extension

Differences Across Methods: 1) EGM approach much faster than VFI as EGM avoids optimization (uses Euler) whereas VFI uses brute-force over all a'. 2) EGM generates a much smoother curve compared to VFI which is kinked because EGM uses continuous interpolation whereas VFI uses discrete grid. 3) EGM is harder to implement (due to the use of the Euler equation and endogenous grid) whereas VFI is easier as it is just maximization.

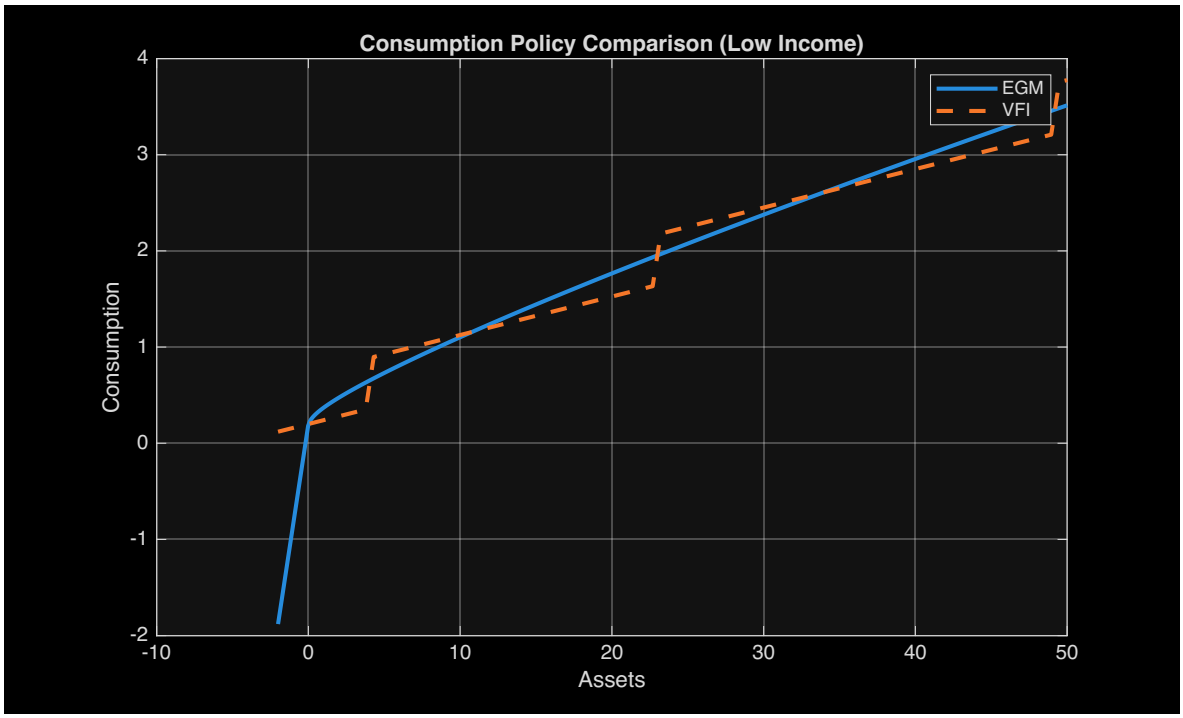

Figure 4: VFI vs. EGM Solution Comparison